

Exploratory Data Analysis (EDA) Report

OF Adult Census Income Analysis

Introduction

Exploratory Data Analysis (EDA) is the process of analyzing a dataset to understand its structure, identify patterns, detect relationships between variables, and discover meaningful insights. It is performed after data cleaning and helps in making informed decisions through statistical analysis and data visualization.

Objective

The objective of Exploratory Data Analysis is to examine the cleaned Adult Census Income dataset, identify important trends and patterns, understand the characteristics of different variables, and determine the factors that influence an individual's income. Visualizations are used to make the analysis easier to understand and interpret.

The Exploratory Data Analysis provided valuable insights into the Adult Census Income dataset through statistical analysis and visualizations. The charts helped in understanding the distribution of income, age, education, occupation, work class, and working hours, as well as the relationships among different variables. These insights can assist organizations and researchers in understanding the factors that influence an individual's income and support better data-driven decision-making.

Visualizations and Analysis

- The following visualizations were created to analyze the dataset:

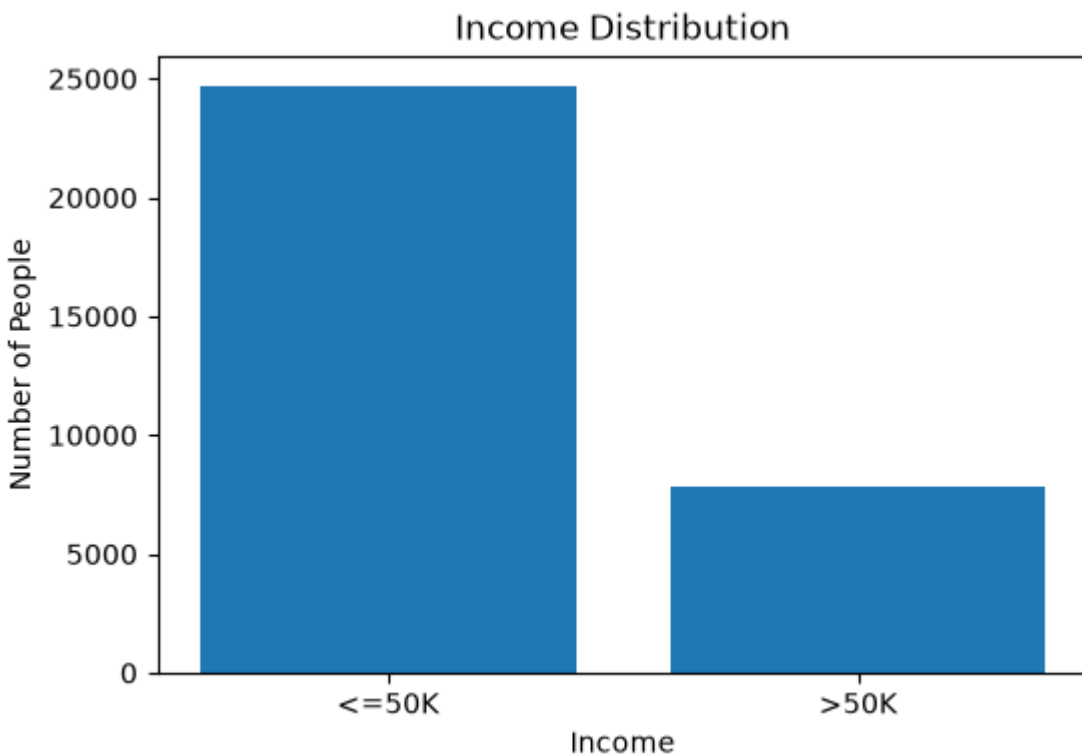
1. Income Distribution (Bar Chart)

Objective: To compare the number of individuals earning $\leq 50K$ and $> 50K$ annually.

Python Code:

```
income_counts = df["income"].value_counts()
plt.figure(figsize=(6,4))
plt.bar(income_counts.index, income_counts.values)
plt.title("Income Distribution")
plt.xlabel("Income")
plt.ylabel("Number of People")
plt.show()
```

Visualization:



Interpretation:

The bar chart shows the distribution of individuals based on their annual income. It helps identify which income category contains the highest number of individuals.

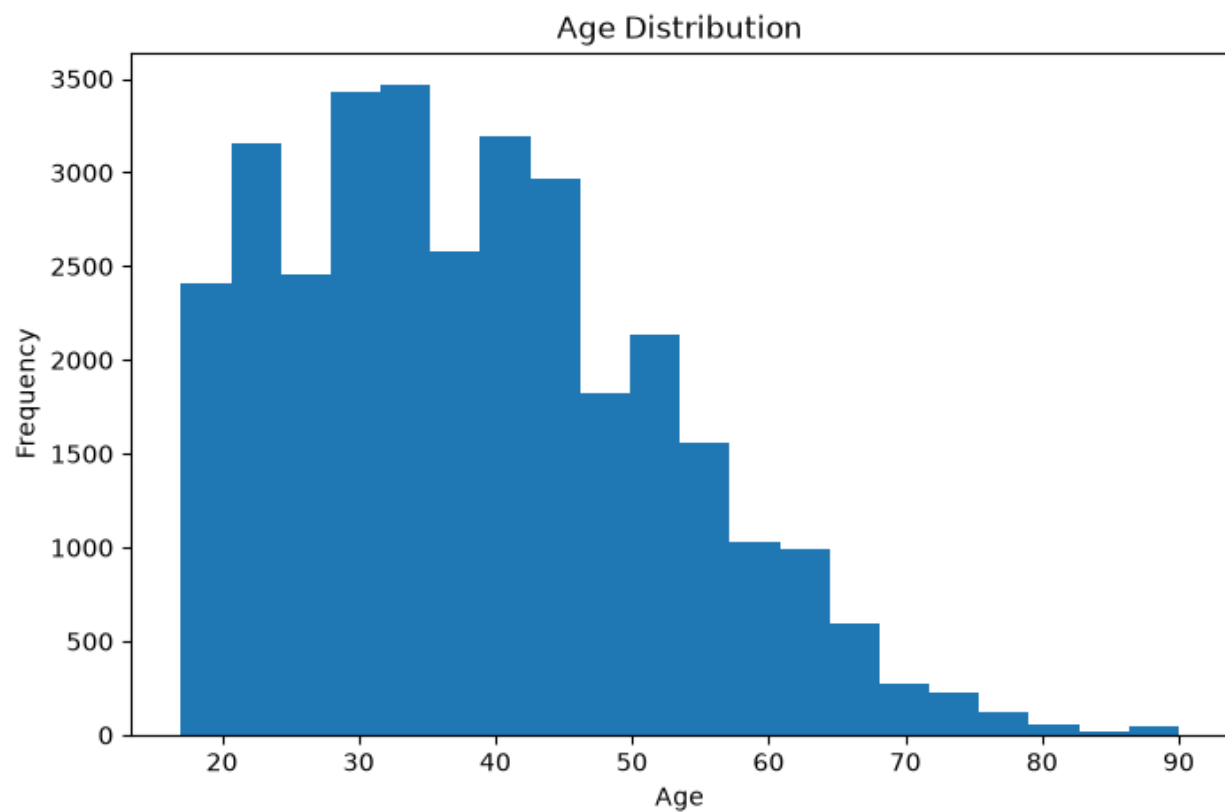
2. Age Distribution (Histogram)

Objective: To study the distribution of age among individuals.

Python Code:

```
plt.figure(figsize=(8,5))
plt.hist(df["age"], bins=20)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```

Visualization:



Interpretation:

The histogram shows the age distribution of individuals and helps identify the most common age groups in the dataset.

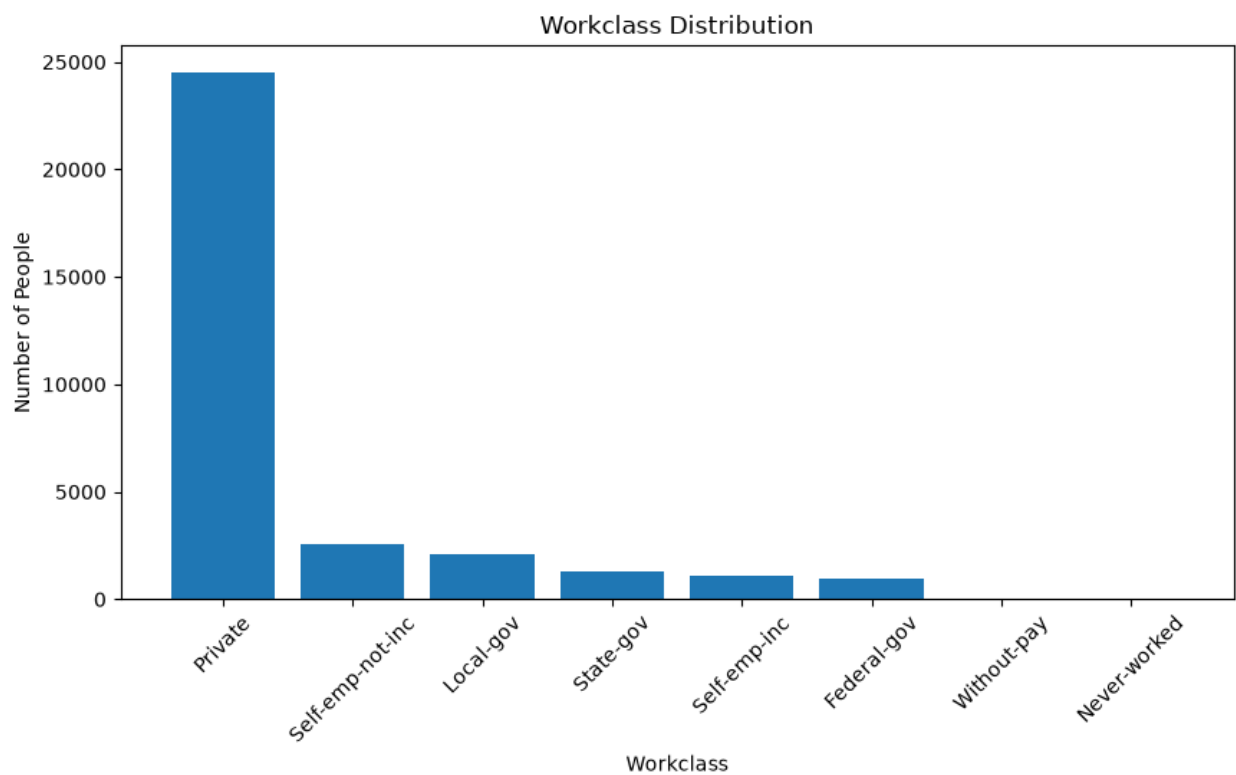
3. Workclass Distribution (Bar Chart)

Objective: To identify the most common work classes in the dataset.

Python Code:

```
workclass_counts = df["workclass"].value_counts()
plt.figure(figsize=(10,5))
plt.bar(workclass_counts.index, workclass_counts.values)
plt.title("Workclass Distribution")
plt.xlabel("Workclass")
plt.ylabel("Number of People")
plt.xticks(rotation=45)
plt.show()
```

Visualization:



Interpretation:

The chart shows the number of individuals belonging to each work class and highlights the most common employment category.

4. Gender Distribution (Pie Chart)

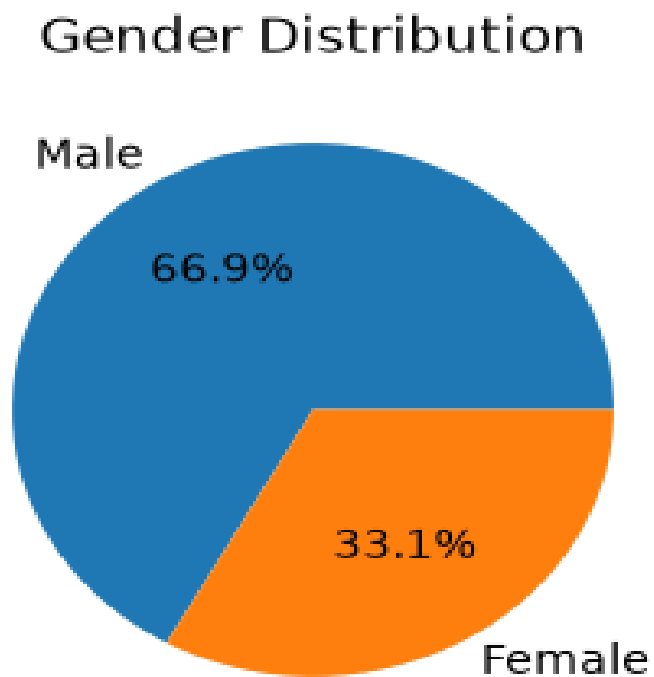
Objective: To compare the proportion of male and female individuals.

Python Code:

```
gender_counts = df["sex"].value_counts()

plt.figure(figsize=(3,3))
plt.pie(gender_counts.values, labels=gender_counts.index, autopct="%1.1f%%")
plt.title("Gender Distribution")
plt.show()
```

Visualization:



Interpretation:

The pie chart illustrates the percentage distribution of males and females in the dataset.

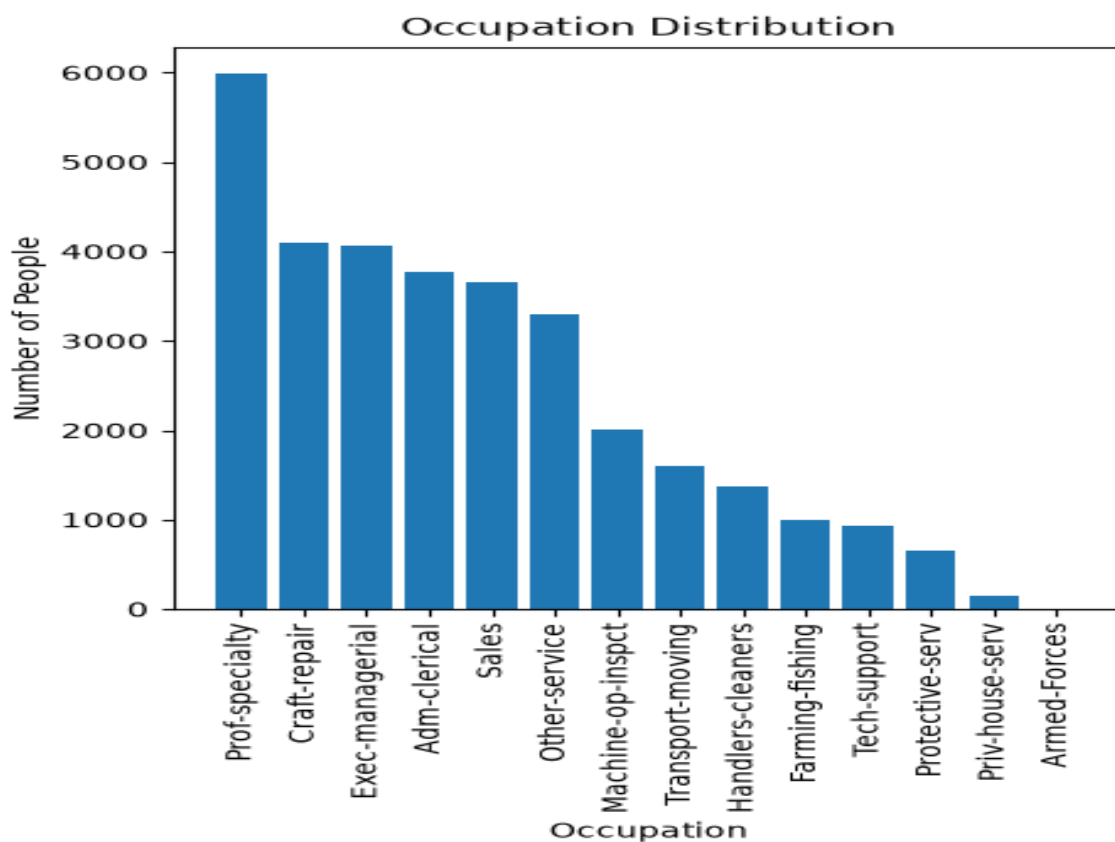
5. Occupation Distribution (Bar Chart)

Objective: To identify the occupations with the highest number of individuals.

Python Code:

```
occupation_counts = df["occupation"].value_counts()
plt.figure(figsize=(5,5))
plt.bar(occupation_counts.index, occupation_counts.values)
plt.title("Occupation Distribution")
plt.xlabel("Occupation")
plt.ylabel("Number of People")
plt.xticks(rotation=90)
plt.show()
```

Visualization:



Interpretation:

The chart shows the frequency of each occupation and helps identify the most common occupations.

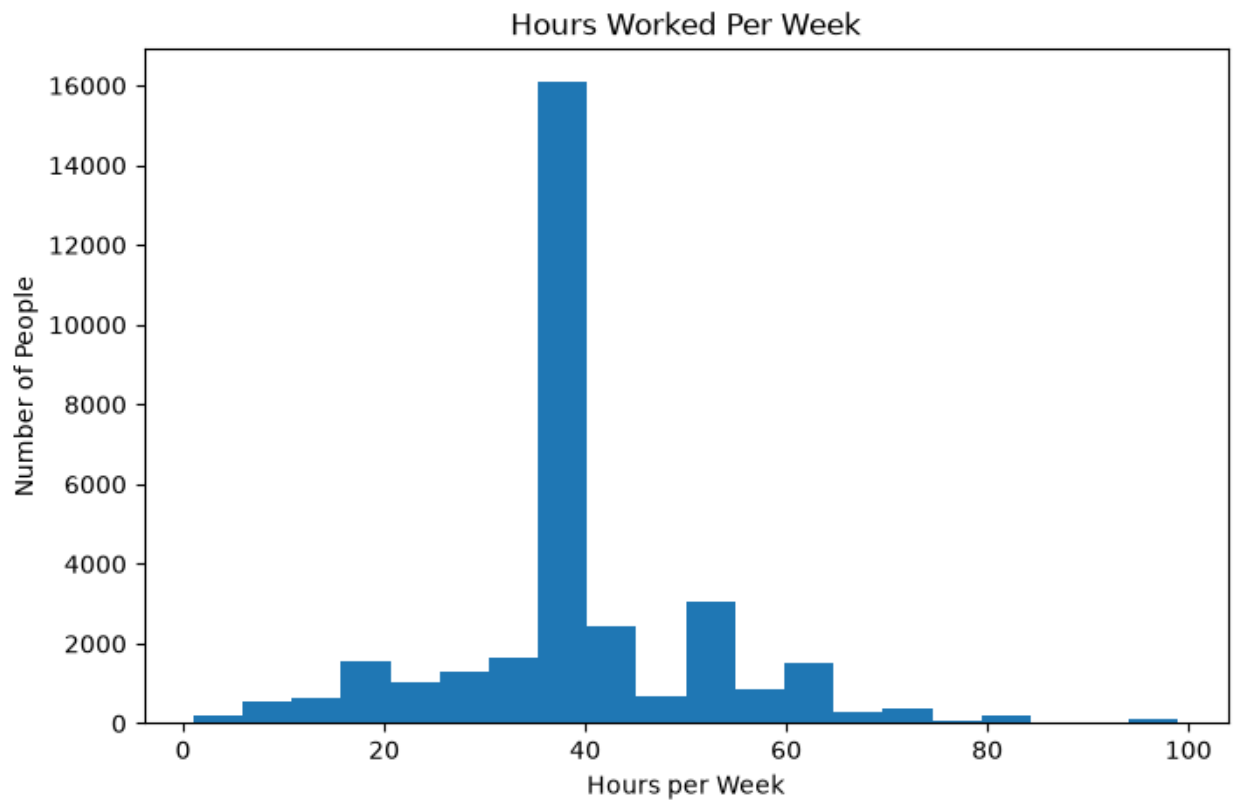
6. Hours per Week Distribution (Histogram)

Objective: To examine the weekly working hours of individuals.

Python Code:

```
plt.figure(figsize=(8,5))
plt.hist(df["hours.per.week"], bins=20)
plt.title("Hours Worked Per Week")
plt.xlabel("Hours per Week")
plt.ylabel("Number of People")
plt.show()
```

Visualization:



Interpretation:

The histogram shows the distribution of working hours and identifies the most common number of hours worked per week.

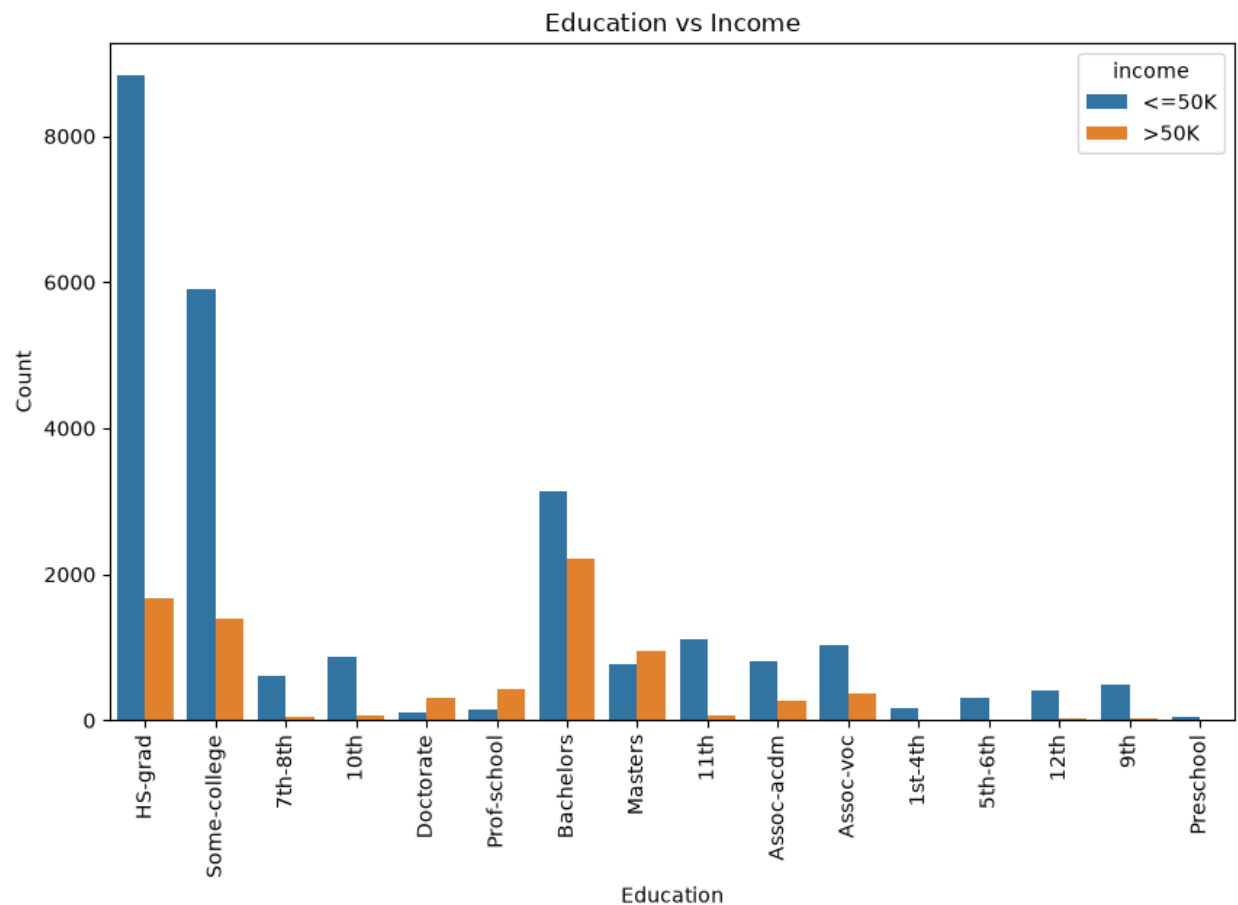
7. Education Level vs Income (Bar Chart)

Objective: To study the relationship between education level and annual income.

Python Code:

```
plt.figure(figsize=(10,6))
sns.countplot(data=df, x="education", hue="income")
plt.title("Education vs Income")
plt.xlabel("Education")
plt.ylabel("Count")
plt.xticks(rotation=90)
plt.show()
```

Visualization:



Interpretation:

The chart helps compare income levels across different education categories and shows how education may influence income.

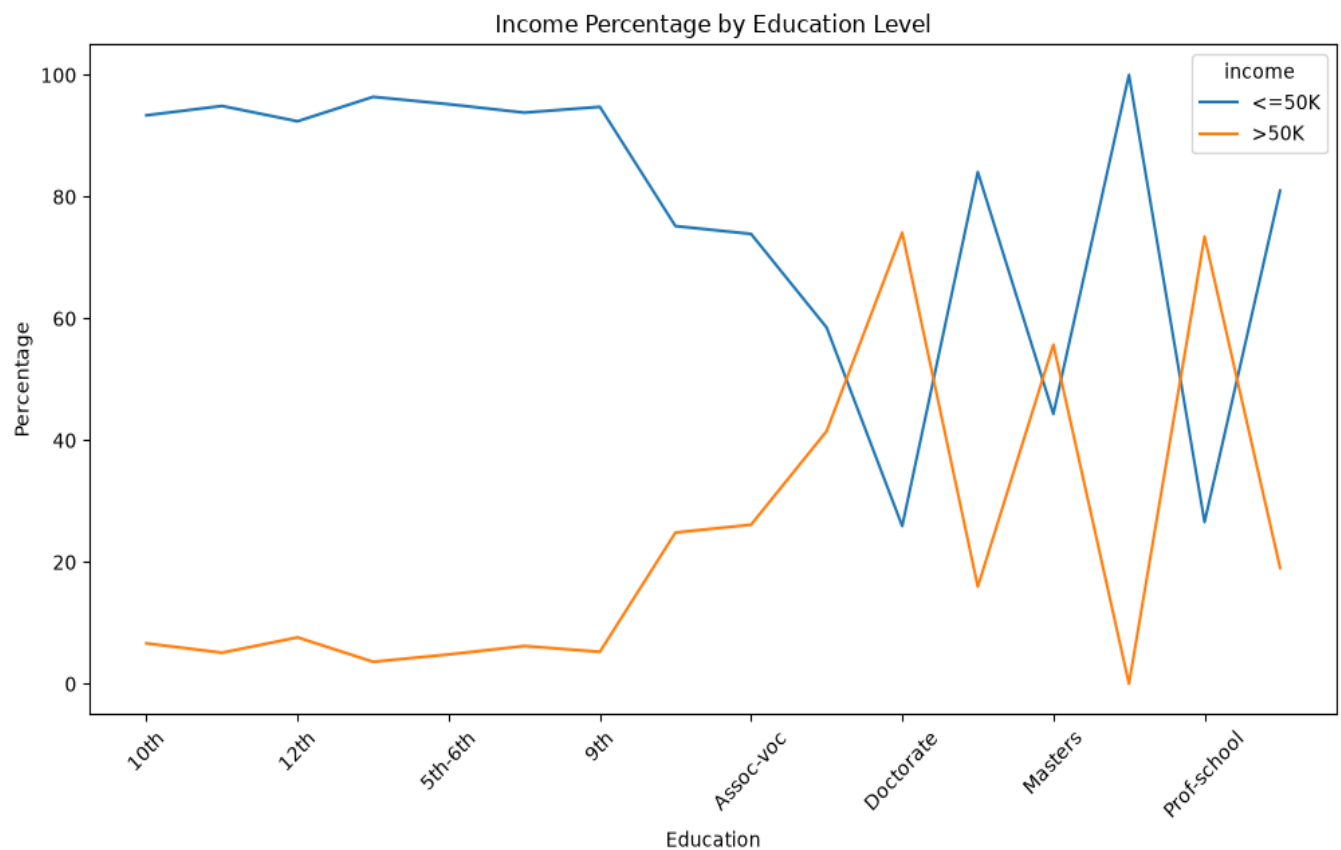
8. Education Level vs Income (Bar Chart)— Percentage

Objective: To compare the percentage of individuals earning $\leq 50K$ and $>50K$ across different education levels.

Python Code:

```
education_income = pd.crosstab(
    df["education"],
    df["income"],
    normalize="index"
) * 100
education_income.plot(figsize=(12,6))
plt.title("Income Percentage by Education Level")
plt.xlabel("Education")
plt.ylabel("Percentage")
plt.xticks(rotation=45)
plt.show()
```

Visualization:



Interpretation:

The visualization compares the percentage of individuals in each income category across different education levels. It shows that income distribution varies across education levels, with higher education levels generally having a greater percentage of individuals earning >50K.

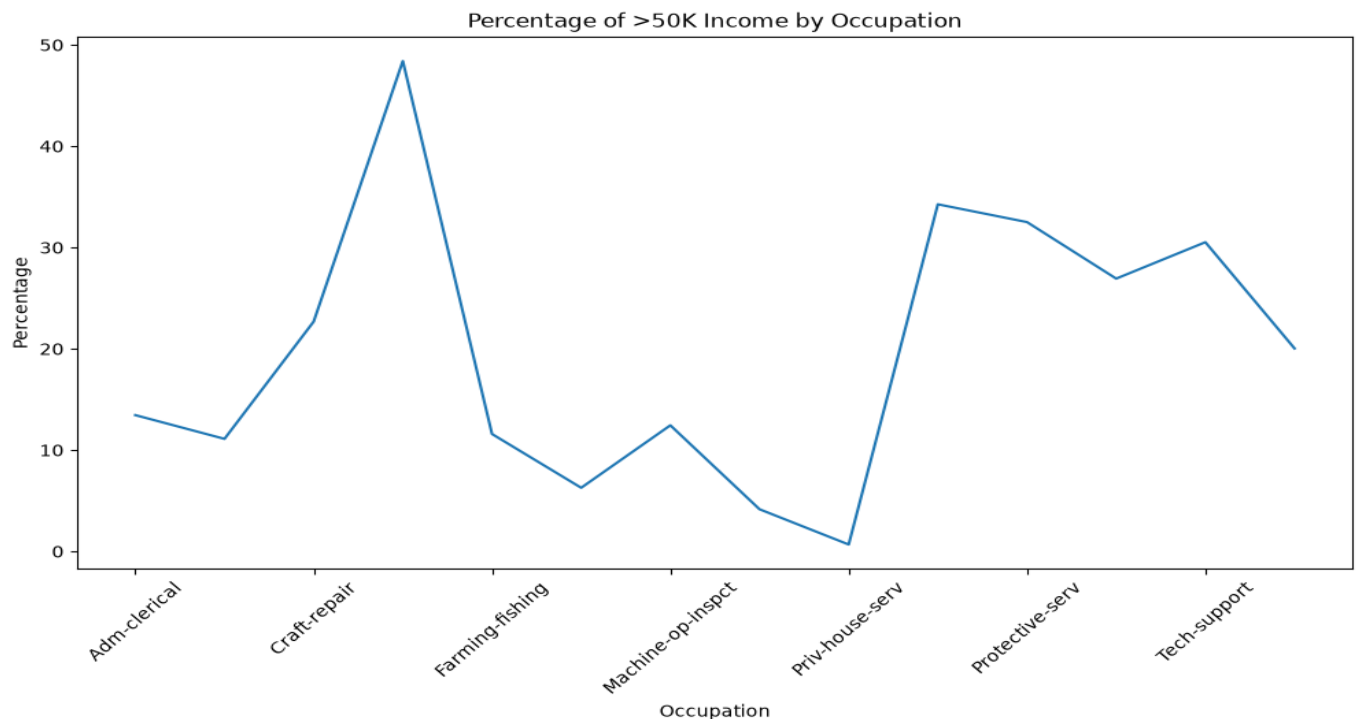
9.Occupation vs Income — Percentage

Objective: To compare the percentage of individuals earning >50K across different occupations.

Python Code:

```
occupation_income = pd.crosstab(
    df["occupation"],
    df["income"],
    normalize="index"
) * 100
occupation_income[ ">50K"].plot(figsize=(12,6))
plt.title("Percentage of >50K Income by Occupation")
plt.xlabel("Occupation")
plt.ylabel("Percentage")
plt.xticks(rotation=45)
plt.show()
```

Visualization:



Interpretation:

The visualization shows that the percentage of individuals earning $>50K$ differs across occupations. Some occupations have a higher percentage of high-income individuals, indicating that occupation is associated with income.

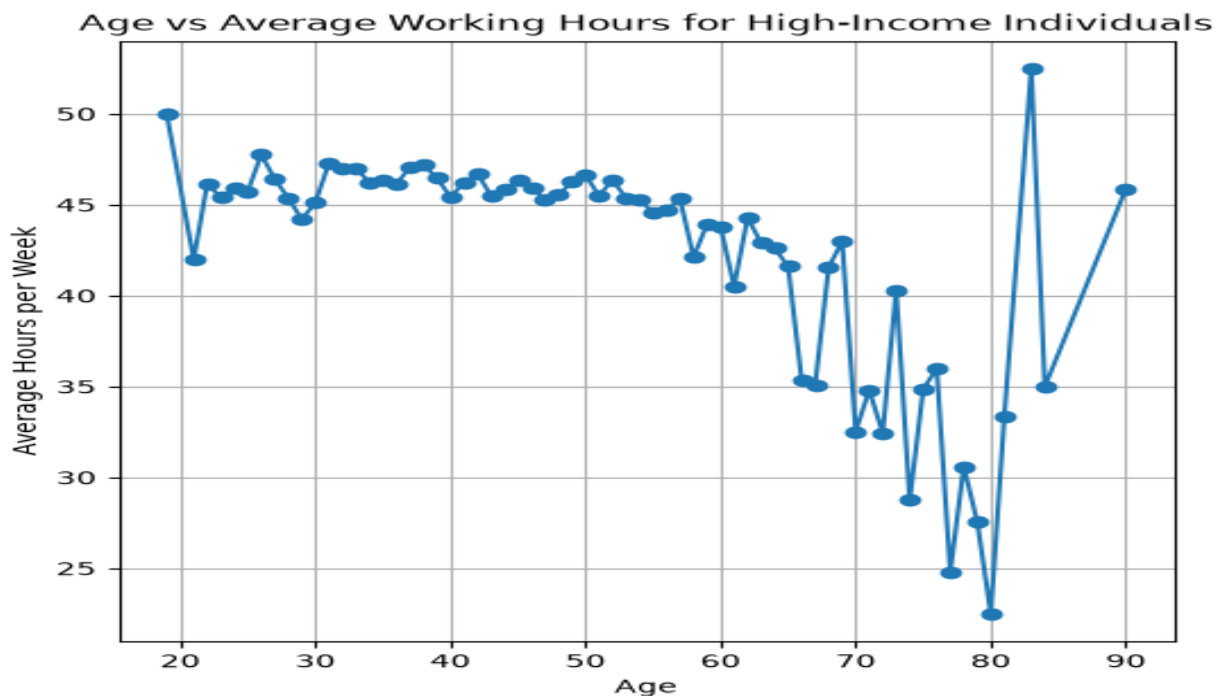
10. Age vs Working Hours for High-Income Individuals

Objective: To analyze the average working hours of individuals earning more than 50K across different age groups.

Python Code:

```
high_income = df[df["income"] == ">50K"]
age_hours = high_income.groupby("age")["hours.per.week"].mean()
plt.figure(figsize=(10,6))
plt.plot(age_hours.index, age_hours.values, marker="o")
plt.title("Age vs Average Working Hours for High-Income Individuals")
plt.xlabel("Age")
plt.ylabel("Average Hours per Week")
plt.grid()
plt.show()
```

Visualization:



Interpretation:

The graph shows how average working hours vary across different age groups among high-income individuals.

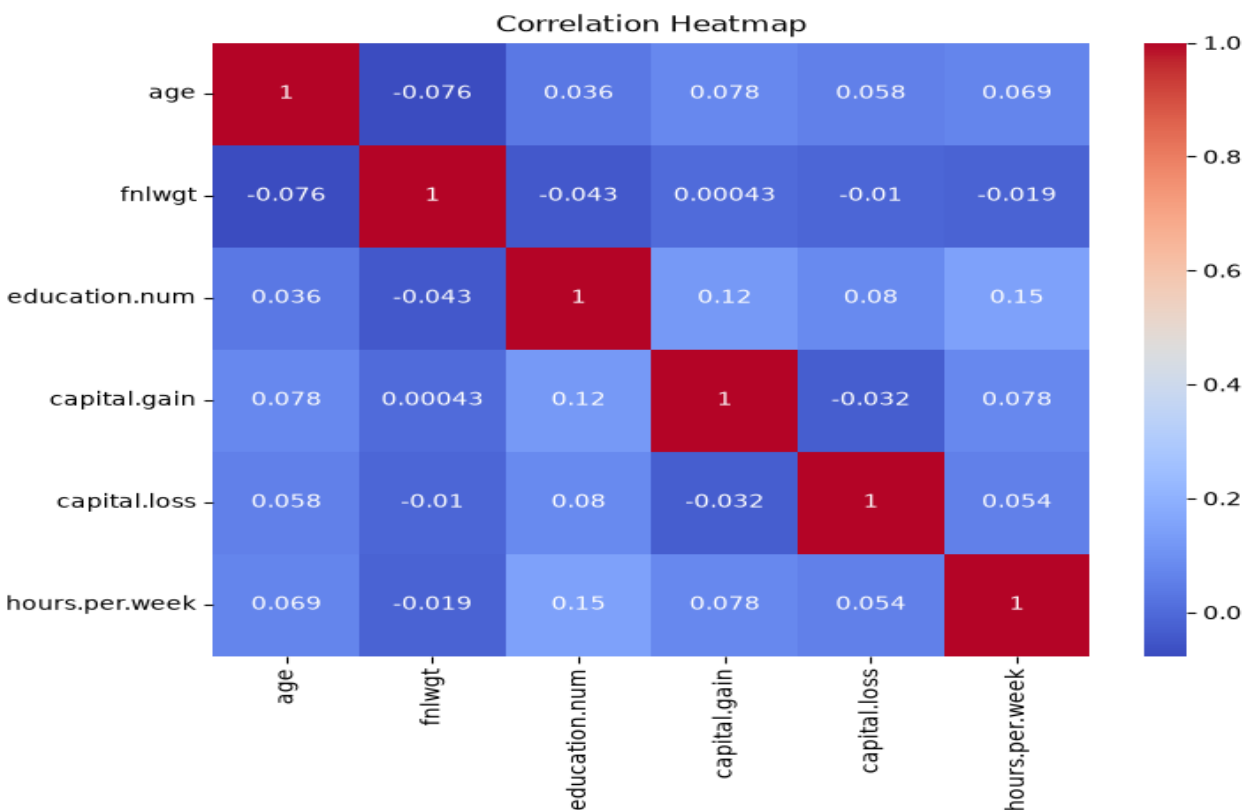
11. Correlation Heatmap

Objective: To analyze the relationship among numerical variables.

Python Code:

```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

Visualization:



Interpretation:

The heatmap shows the correlation between numerical variables and helps identify positive or negative relationships.

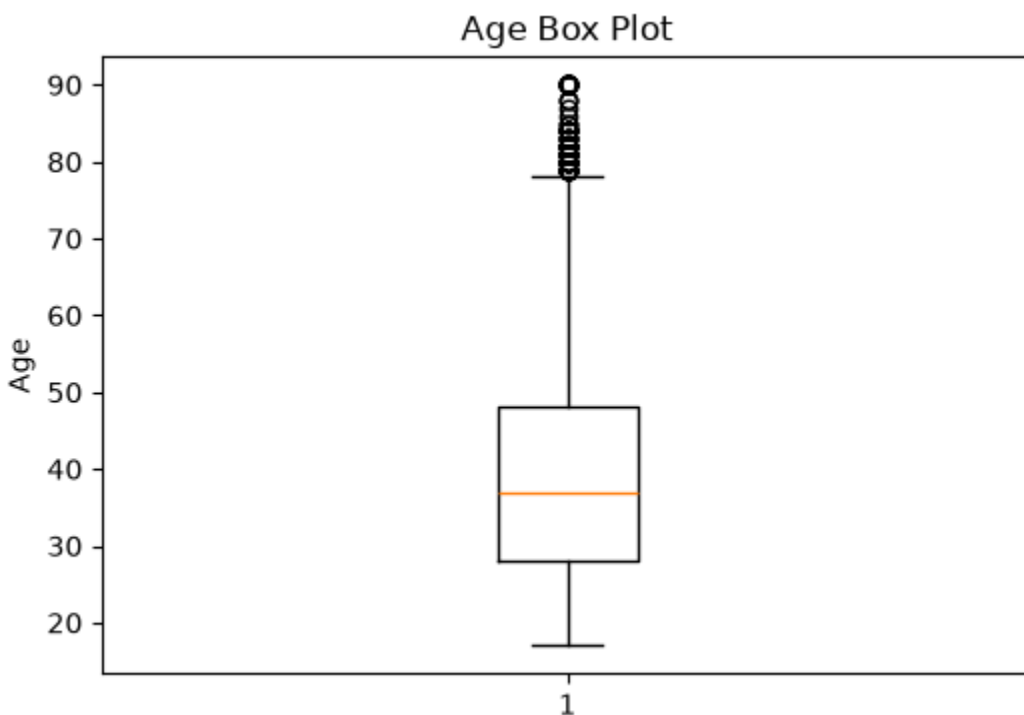
12. Age Outlier Analysis (Box Plot)

Objective: To detect outliers in the age column.

Python Code:

```
plt.figure(figsize=(6,4))  
plt.boxplot(df["age"])  
  
plt.title("Age Box Plot")  
plt.ylabel("Age")  
plt.show()
```

Visualization:



Interpretation:

The box plot helps identify unusual age values (outliers) and shows the spread of the age data.

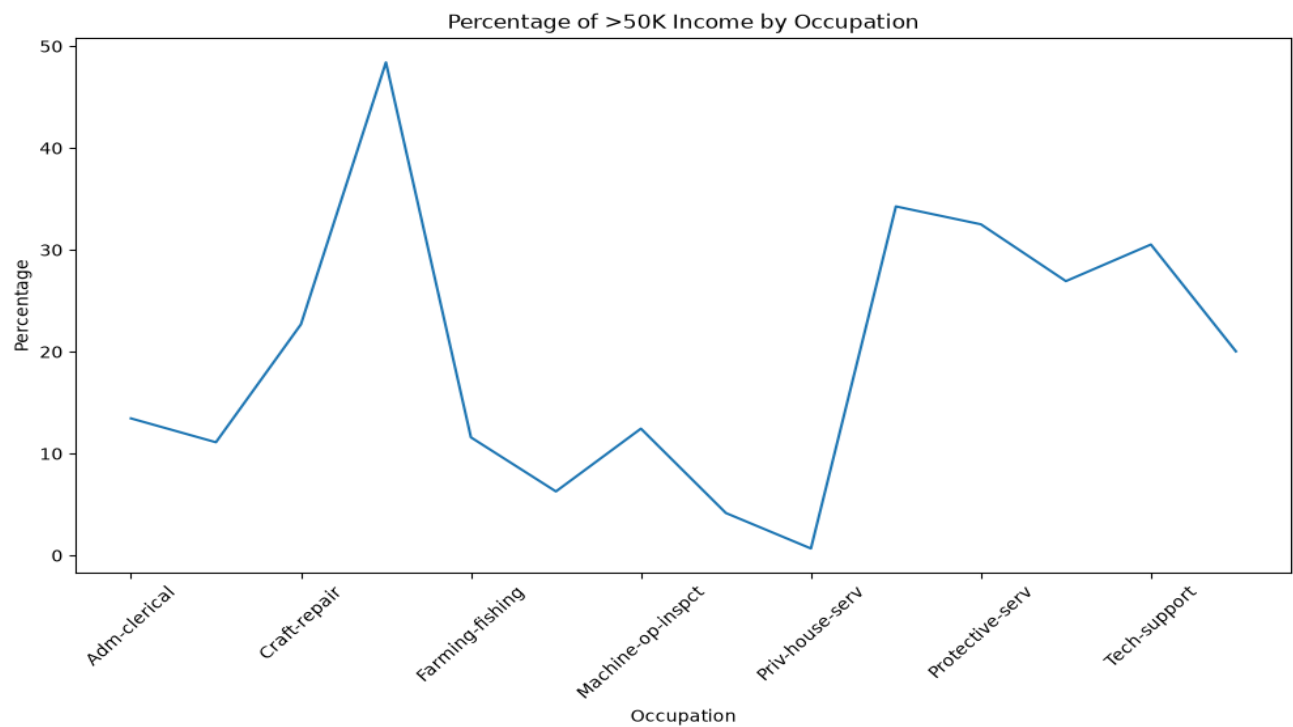
13.Occupation vs Income — Percentage

Objective: To compare the percentage of individuals earning >50K across different occupations.

Python Code:

```
occupation_income = pd.crosstab(  
    df["occupation"],  
    df["income"],  
    normalize="index"  
) * 100  
  
occupation_income[ ">50K"].plot(figsize=(12,6))  
  
plt.title("Percentage of >50K Income by Occupation")  
plt.xlabel("Occupation")  
plt.ylabel("Percentage")  
plt.xticks(rotation=45)  
plt.show()
```

Visualization:



Key Findings

The EDA revealed several important patterns in the Adult Census Income dataset:

- The majority of individuals belong to the **<=50K income category**, while a smaller proportion belongs to the **>50K** category.
- The **Private** workclass represents the largest group of individuals in the dataset.
- The age of individuals ranges from **17 to 90 years**, with most individuals falling within the working-age population.
- **40 hours per week** is the most common number of working hours.
- **HS-grad** is the most common education level, followed by **Some-college** and **Bachelors**.
- The **Education vs Income** visualization indicates that education level is associated with income, with higher education levels generally showing a greater number of individuals in the **>50K** category.
- Most individuals have **zero capital gain and zero capital loss**, indicating that investment-related gains and losses are uncommon in the dataset.
- The correlation heatmap helped identify relationships among numerical variables, while the box plot provided an overview of the distribution and possible outliers in age.

Conclusion

The Exploratory Data Analysis provided a clear understanding of the major characteristics and patterns within the Adult Census Income dataset. Through statistical analysis and visualizations, the study examined income distribution, age, education, workclass, occupation, and working hours, as well as relationships between selected variables.

Overall, the analysis indicates that **education, occupation, workclass, age, and working hours are useful factors for understanding income patterns**. The visualizations made these patterns easier to identify and interpret, providing meaningful insights from the cleaned dataset and establishing a foundation for further analysis.

